

Ex. 1. What is the variance of a random variable that is Bernoulli or binomially distributed?

Ex. 2. If X and Y are independent then is it true that X and $X+Y$ are independent as well?

Ex. 3. Prove that $E(X - a)^2 = \text{Var } X + (a - EX)^2$.

Ex. 4. Let X be a random variable having expected value μ and variance σ^2 . Find the expected value and variance of

$$Y = \frac{X - \mu}{\sigma}.$$

Ex. 5. If X is a binomial random variable with expected value 6 and variance 2.4, find $P(X = 5)$.

Ex. 6. Find the variance of a random variable that is uniformly distributed on (a, b) .

Ex. 7. Consider n rolls of a die. We have seen that, if X_j is the outcome if the j th roll, then $E(X_j) = 7/2$ and $V(X_j) = 35/12$. Thus, if S_n is the sum of the outcomes, and $A_n = S_n/n$ is the average of the outcomes, we have $E(A_n) = 7/2$ and $V(A_n) = (35/12)/n$. Therefore, as n increases, the expected value of the average remains constant, but the variance tends to 0. If the variance is a measure of the expected deviation from the mean this would indicate that, for large n , we can expect the average to be very near the expected value

Ex. 8. A number is chosen at random from the set $S = \{-1, 0, 1\}$. Let X be the number chosen. Find the expected value, variance, and standard deviation of X .

Ex. 9. X is a random variable with $E(X) = 100$ and $V(X) = 15$. Find

(a) $E(X^2)$.

(b) $E(3X + 10)$.

(c) $E(-X)$.

(d) $\text{Var}(-X)$.

(e) $\text{SD}(-X)$.

$$\begin{aligned} \text{Var}(X_1 - X_2) &= \text{Var}(X_1 + (-X_2)) = \text{Var } X_1 + \text{Var}(-X_2) = \\ &= \text{Var } X_1 + \text{Var } X_2 \cdot (-1)^2. \end{aligned}$$

Ex 1 $X \sim \text{Bern}(p)$. $EX = 1 \cdot p + 0 \cdot (1-p) = p$

